Sparse-Autoencoder Dissection of Emotion Features in Frozen Speech Encoders

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Abstract

Linear probes tell us that emotion becomes *readable* in the middle-to-late layers of frozen speech transformers — but not *which* internal features carry it, and not whether a probe is listening to *how* something is said (acoustics) or *what* is said (words). We open these models up with Sparse Autoencoders (SAEs): we train a TopK SAE on the per-frame activations of six frozen large encoders (HuBERT, WavLM, wav2vec2, w2v-BERT, MERT, Whisper) and read off the individual features it learns. The trick that makes this work is a lexicon-controlled corpus, CREMA-D (91 actors × 12 *fixed* sentences × 6 emotions), plus a confound check that keeps a feature only if it tells us more about emotion than about the sentence or the speaker — otherwise the SAE just rediscovers phonemes. Four findings follow: (i) emotion is *organised* into clean features earlier (layers 2–7) than where a probe reads it best (layer 10); (ii) all six encoders share a common, speaker-invariant acoustic feature basis; (iii) the recovered signal is acoustic, not lexical; and (iv) on incongruent speech (EMIS), the "lexical shortcut" is a *deep-layer* habit that only the ASR-trained encoders pick up, while Whisper stays grounded in the voice at every layer.

Keywords

speech emotion recognition, sparse autoencoders, interpretability, disentanglement, acoustic features

1 Why this study

Self-supervised speech encoders sit on top of the speech-emotion leaderboards, but they were originally trained to transcribe words, so they carry a lot of linguistic information. That raises an uncomfortable question: when a probe reads emotion out of one of these models, is it hearing the *tone of voice*, or is it quietly reading the *words*? A model that leans on the words will fail exactly where we most need it — sarcasm, accents, and languages where the lexicon is unfamiliar.

Standard probing cannot answer this. It tells us *where* emotion is decodable, but treats the layer as a black box. We instead use a Sparse Autoencoder, which factors the dense activation of a layer into a large dictionary of sparse, mostly single-meaning "features." Inspecting those features lets us ask *which* ones track emotion, *what* they respond to (prosody vs. words), and *whether* they are shared across models and speakers. In short, we move from **decodability** to **mechanism**.

A known hazard is that an SAE trained on ordinary speech mostly rediscovers *phonemes* [4, 9]. We sidestep it by anchoring on CREMA-D, where the 12 carrier sentences are identical across emotions, so a recovered emotion signal cannot be coming from the words. This doubles as a built-in sanity check.

What we contribute. (1) a confound-controlled recipe for finding genuine emotion features in speech SAEs; (2) the observation that disentanglement and decodability peak at *different* depths; (3) an acoustic-vs-lexical label per feature plus a causal sufficiency test; and (4) a depth-resolved picture of the lexical shortcut on EMIS. Code and artifacts are released.

Table 1: What we reuse from the probing project, and what is new here.

From the probing project	In this work
6 frozen encoders (`feature_extractors`)	same encoders, read un-pooled
mean-pooled hidden states, 25 layers	same layers, kept *per-frame*
logistic probe (scaler, 5-fold CV)	same probe: depth curve + ablation
cached embeddings (EXP folders)	inventory + accuracy reference
CREMA-D, MSP-Podcast, IEMOCAP	same; plus EMIS (added)
behavioral findings (depth, LOSO, EN↔ZH, EMIS text-bias)	the results we explain mechanistically

2 Method

2.1 Where we start: the probing project

This work is the direct continuation of an earlier probing study on the same six encoders. That study used one shared pipeline: `TransformerExtractor` runs each *frozen* encoder and *mean-pools* every layer's hidden state into one vector per clip (25 vectors per clip: a CNN front-end plus 24 transformer layers); `probe_all_layers` then fits a logistic-regression probe per layer (StandardScaler, 5-fold cross-validation). From that it established *where* emotion is decodable and produced four behavioral findings we take as targets: a middle-to-late accuracy peak by depth, the EN↔ZH cross-lingual collapse, the MESD leave-one-speaker-out drop, and the per-encoder EMIS text-bias (Whisper resists, the SSL encoders do not). Those findings say *what* happens, not *why*.

We reuse that pipeline almost verbatim and add one thing: instead of probing the pooled vector, we factor the activation into interpretable features with an SAE. Table 1 is the exact mapping.

2.2 The SAE pipeline

Figure 1 shows the whole method at a glance; we describe each stage below.

(1) Per-frame activations. The probe averages each layer over time, giving one vector per clip — far too few examples to fit a large SAE. So we re-run the same frozen encoders and keep the activations *un-pooled*: one vector per audio frame, per layer (Whisper's padding frames masked). CREMA-D layer 13 alone gives 940,273 frames.

(2) Train the SAE. A TopK SAE (d_{in}=1024, dictionary 8192, k=32 active features per frame) with a unit-norm decoder and an auxiliary loss that revives dead features. It is a faithful compressor: 93% of the activation variance explained, essentially no dead features (Table 4).

(3) Find emotion features (a triple filter). A feature counts as an emotion feature only if it passes all three: (a) *significant* — its firing

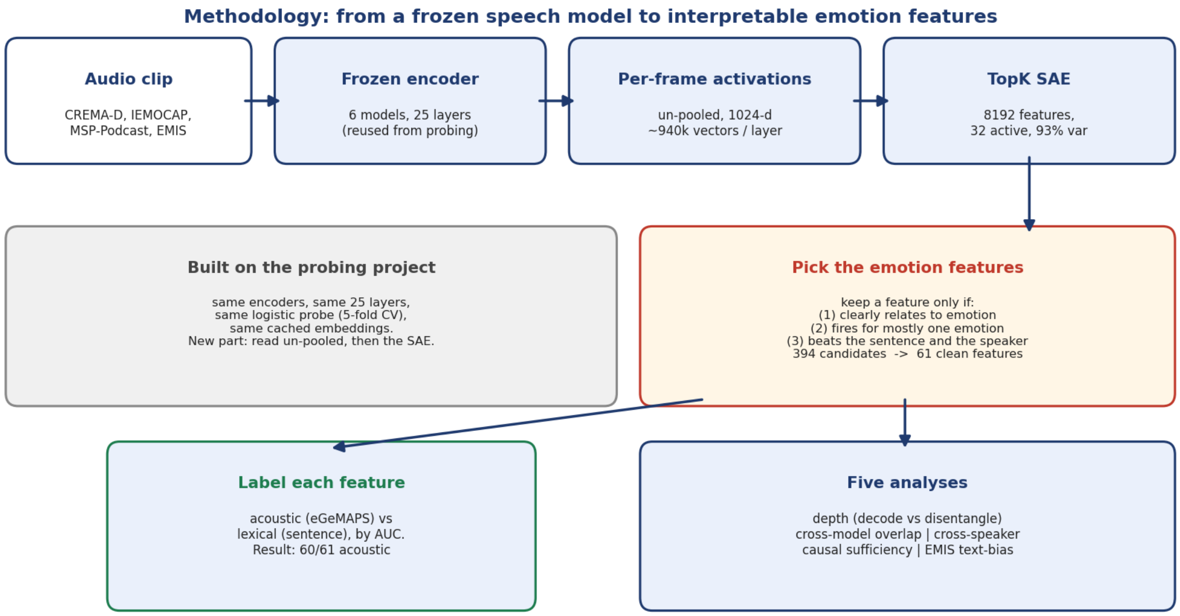

Figure 1: The method on one page. We reuse the probing project's frozen encoders, layers, probe, and cached embeddings; the new part (top-right) reads each layer un-pooled and factors it with an SAE, then keeps only features that pass three filters, labels them acoustic or lexical, and runs five analyses.

carries real mutual information with the emotion label (5σ over a shuffle null); (b) *monosemantic* — at least half its activity falls on one emotion; and (c) it beats the *confounds* — it tells us more about emotion than about the spoken sentence (a stand-in for phonetics) or the speaker. Filters (a)+(b) alone flag 394 features, but two-thirds are really about the sentence or the speaker; the full filter leaves **61** clean emotion features, dominated by anger and sadness.

(4) Label each feature acoustic or lexical. We ask whether a feature's firing is better predicted by acoustic descriptors (eGeMAPS [6]) or by the sentence identity (lexical content), using classification AUC (the features are ~96% silent, so R^2 is unusable).

(5) Five analyses on these features. Depth (count features at all 25 layers, vs. the probe curve); cross-encoder (match features across the six models); cross-speaker (leave-one-speaker-out over 91 actors); causal (re-probe emotion from only-acoustic vs. only-lexical feature subsets, reusing the project's probe); and EMIS (train on congruent, test on incongruent speech, by layer).

3 What we found

3.1 Emotion is organised earlier than it is decoded

If you just train a probe, accuracy climbs and peaks *deep* (layer 10) and stays high. But the *clean* emotion features tell a different story: their count peaks *early* (layer 7), and their purity even earlier (layer 2). By the deep layers, emotion is still easy to decode but has smeared out across many directions — only 24–47 clean features remain (Fig. 2, top-left; Fig. 3, left). In plain terms: **the model tidies emotion into neat features early, then a probe reads it best later, once it has been re-mixed for linear separability.** Decodability and disentanglement are not the same thing.

3.2 The same acoustic features, across encoders and speakers

Every encoder ends up with a healthy set of emotion features (Table 2), all anger/sadness-dominated. When we match features across encoders by how they fire, Whisper overlaps with the self-supervised models just as much as they overlap with each other (0.40 vs. 0.39). So on fixed-lexicon speech, *Whisper is not special* — its much-discussed difference is about text, which CREMA-D cannot expose (we return to it in §3.4). The features also travel across people: under a strict leave-one-speaker-out test over all 91 actors, accuracy barely moves (drop ≈ 0) and each feature fires for ~95% of the actors who express its emotion. This also explains a known result elsewhere: the large speaker-out drop on MESD (~0.30) comes from *speaker-entangled* features — precisely the ones our confound check throws away.

3.3 The signal is acoustic, not lexical

On fixed-lexicon CREMA-D, an emotion feature *must* be acoustic — and 60 of 61 are (mean acoustic AUC 0.83 vs. 0.66 for sentence identity; Fig. 2, top-middle). The causal test agrees: if we let emotion be decoded only from the acoustic features we get 0.55, but only from the lexical features just 0.43 — barely above the 0.42 we get

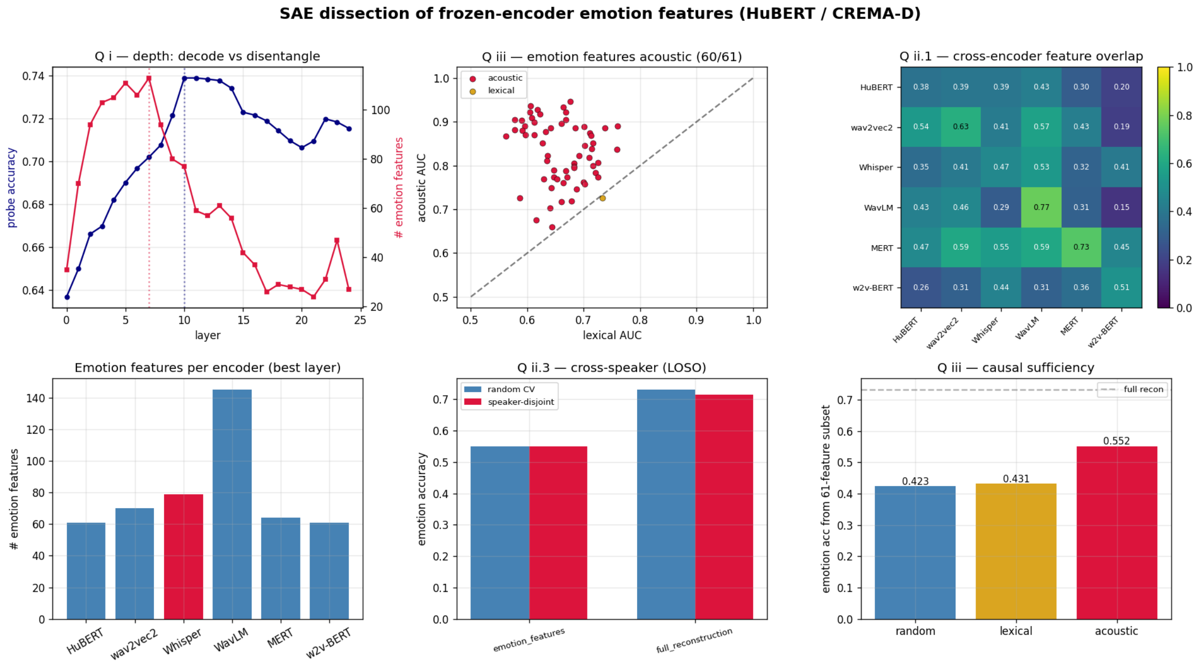

Figure 2: The whole study at a glance (HuBERT / CREMA-D). Top: depth — probe accuracy vs. number of clean emotion features per layer; each feature acoustic vs. lexical (above the line = acoustic); how features overlap across the six encoders. Bottom: number of emotion features per encoder; cross-speaker test (random vs. leave-speaker-out); and the causal test (how well emotion decodes from only-acoustic vs. only-lexical vs. random features).

from random features (Fig. 2, bottom-right). The acoustic features are *sufficient*; the lexical ones are not.

3.4 The lexical shortcut is a deep-layer habit

CREMA-D cannot test the text shortcut, because its text is fixed. EMIS [5] can: it is synthetic speech where the *written* emotion and the *spoken* emotion are deliberately mismatched. We train a probe on the matched clips and test on the mismatched ones; "text-bias" is how often the model goes with the words instead of the voice. The result (Table 3, Fig. 4): every encoder is grounded in the voice early on, but the three ASR-trained self-supervised models (HuBERT, WavLM, wav2vec2) *develop* the shortcut in their deep layers. Whisper never does — it stays voice-grounded at every layer — and the music model MERT and w2v-BERT never take it either. This reproduces the previously published per-encoder text-bias and adds the new part of the story: the shortcut lives at a depth Whisper simply never reaches.

4 The pipeline, and what is released

The study runs as nine numbered, reproducible steps (Table 5), each a module under `src/sae/steps/` callable as `python -m src.sae.steps <step>`. Heavy jobs (extraction, the 25-layer SAE grid) fan out as SLURM array jobs on a single V100 each. Everything needed to reproduce a number is versioned: the per-step CSVs, plots, and metrics live under `results/SAE/`; the large activation tensors and SAE checkpoints live on scratch (too big for version control) with pointers recorded. CREMA-D was pulled from a public mirror and EMIS from its open Zenodo release.

Table 2: Per-encoder emotion features at each encoder's best CREMA-D layer.

Encoder	Best layer	# emotion feats	Family
HuBERT	13	61	SSL (ASR-style)
wav2vec2	10	70	SSL (ASR-style)
Whisper	12	79	Supervised ASR
WavLM	14	145	SSL (ASR-style)
MERT	9	64	Music SSL
w2v-BERT	17	61	SSL (multilingual)

Table 3: EMIS text-bias (text − audio accuracy) at each encoder's most text-biased layer; positive = takes the lexical shortcut. Raw mean-pooled activations.

Encoder	Text-bias	Verdict
wav2vec2 (L15)	+0.87	takes the shortcut
HuBERT (L20)	+0.79	takes the shortcut
WavLM (L20)	+0.73	takes the shortcut
Whisper (L24)	**−0.11**	**resists (voice-grounded)**
MERT	−0.60	resists (music model)
w2v-BERT	−0.60	resists

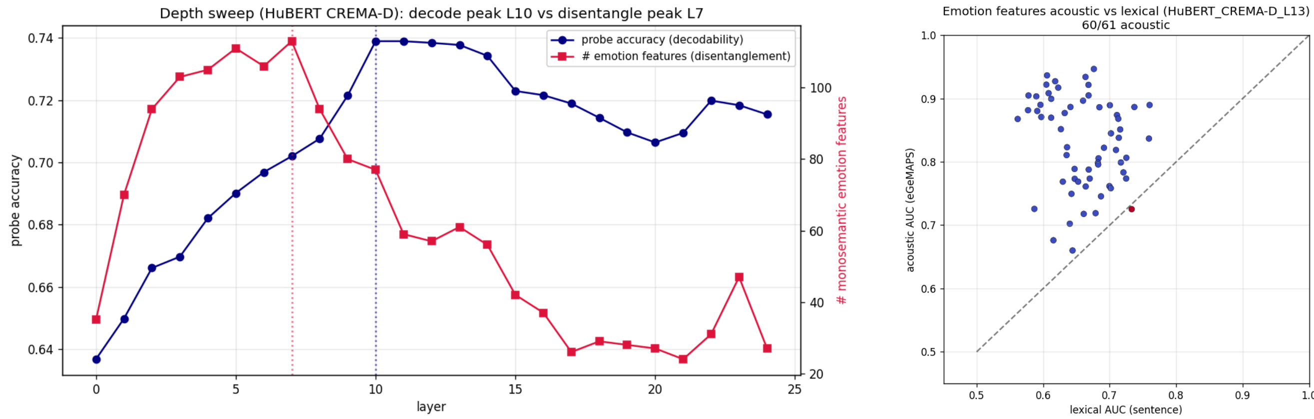

Figure 3: Left: depth — a probe reads emotion best deep (L10) while the count of clean emotion features peaks early (L7). Right: each emotion feature scored on acoustic (eGeMAPS) vs. lexical (sentence) prediction — nearly all sit on the acoustic side of the diagonal.

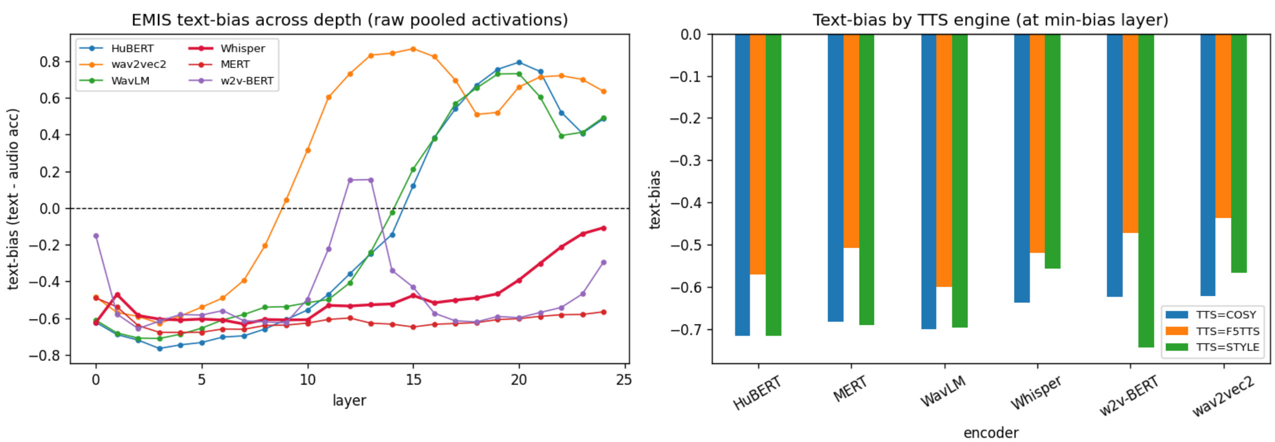

Figure 4: EMIS text-bias (text − audio accuracy) across all 25 layers. The "use-the-words" shortcut appears only in the *deep* layers of the ASR-trained encoders (HuBERT/WavLM/wav2vec2); Whisper (red) stays voice-grounded at every layer, and MERT/w2v-BERT never take it. Right panel: the same broken down by TTS engine.

Table 4: Key numbers (HuBERT / CREMA-D, layer 13 unless noted).

Quantity	Value
SAE reconstruction (variance explained)	93% (FVU 0.07)
Dictionary / sparsity / dead features	8192 / k=32 / 4
Emotion feats: significant → clean → confound-OK	6407 → 394 → 61
Decode peak / disentangle peak / purity peak	L10 / L7 / L2
Acoustic features (positive control)	60/61
Decode from acoustic / lexical / random feats	0.55/0.43/0.42
Cross-speaker leave-one-out drop	≈ 0
EMIS: Whisper L24 / mean ASR-SSL text-bias	−0.11/+0.80

Table 5: The pipeline. Each step is one reproducible command.

Step	Does	Headline output
0	inventory the cached embeddings	coverage map
0.5	extract per-frame activations	940k frames
1	train a TopK SAE	93% variance, 4 dead
2	find emotion features (confound-ctrl)	394 → 61
3	label acoustic vs. lexical	60/61 acoustic
4	depth sweep (all 25 layers)	L10 vs. L7
5	cross-encoder overlap	Whisper not special
6	cross-speaker (leave-one-out)	drop ≈ 0
7	causal sufficiency	0.55 vs. 0.43
8	EMIS text-bias by depth	shortcut is deep

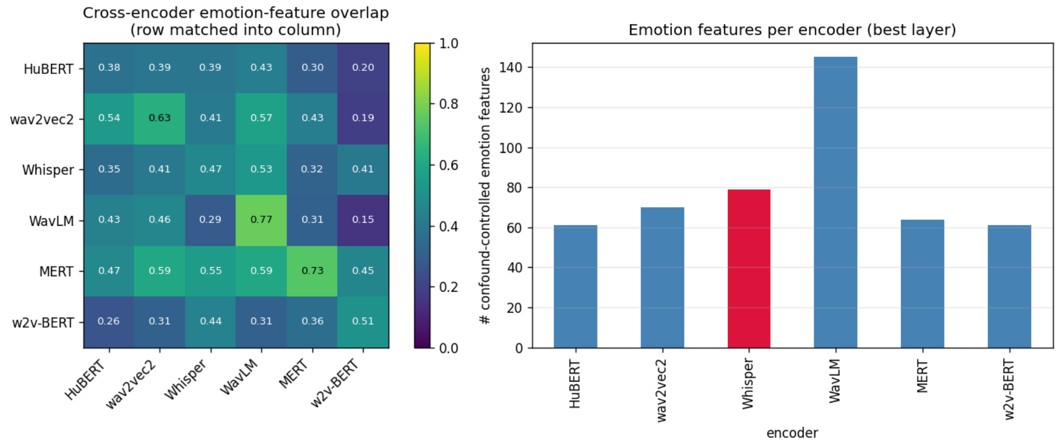

Figure 5: Cross-encoder picture. Left: how much each encoder's emotion features overlap with every other's — Whisper is not an outlier. Right: number of clean emotion features per encoder at its best layer.

5 Related work and how we differ

The closest method paper is *AudioSAE* [3] (EACL 2026): the same SAE recipe (TopK, 8×, per-frame, all layers) on two encoders, even with a cross-model feature-overlap analysis. But it treats emotion as just one of four generic probe tasks — no dedicated emotion-feature interpretation, no confound check, no lexicon-controlled corpus, and no acoustic-vs-lexical split. Two further SAE-for-speech papers find phonetic features [9] or singing-technique features [8], not affect, and do not sweep depth. The closest mechanistic SER paper [7] uses probing, logit-lens and CKA but *no* SAEs; the acoustic-vs-lexical question itself is studied with text-dependency analyses [1, 2] and the EMIS benchmark [5]. Our specific combination — confound-controlled emotion features on a lexicon-controlled factorial corpus across six encoders, plus the depth and EMIS results — is, to our knowledge, new.

6 Limitations and next steps

Because emotion is spread across thousands of features, removing a handful barely dents accuracy, so we rely on the *sufficiency* test rather than ablation. One caution we learned the hard way: the SAE's *reconstruction* must not stand in for raw activations in the EMIS probe — doing so spuriously flips Whisper. Most depth results are on HuBERT/CREMA-D; the cross-encoder and EMIS arms use one layer per encoder rather than full 25-layer grids. Two follow-ups are coded and waiting on data: a causal ablation on IEMOCAP with *real* transcripts, and a SpeechCraft cross-lingual study to explain the published English↔Chinese collapse as disjoint feature sets. Beyond those: feature *steering* (turning an emotion feature up to shift predictions) would upgrade our correlational claims to causal ones.

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